

Phi-Renormalization for Axisymmetric Navier–Stokes with Swirl

Algebraic Cancellation of the $1/r^4$ Axis Term

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Disclaimer. This note treats the axisymmetric-with-swirl class and a Q_1 -augmented / Phi-renormalized system. Classical regularity without augmentation remains open; unconditional 3D Navier–Stokes is not claimed.

Proof Status Dashboard

All results are for the axisymmetric-with-swirl class unless stated otherwise.

Result	Status	Location
Substitution $\Gamma = ru_\theta$, $\Phi = \Gamma/r^2$	Done	§2
Phi-Renormalization: $1/r^4$ cancels algebraically	Proved	Thm 2.2
No Hardy inequality required	Yes	Thm 2.2
Global C^∞ for augmented system	Proved	Thm 3.2
Gronwall-free convergence at rate $\varepsilon^{4/(\beta+2)}$	Proved	Thm 4.2
Classical system: global regularity (no augmentation)	Open	Open Problem
Classical 3D Navier–Stokes	Not claimed	—

Abstract

We study the 3D incompressible Navier–Stokes equations for axisymmetric-with-swirl data. The central difficulty is the $1/r^4$ centrifugal forcing singularity on the axis $r = 0$, which resists standard energy methods and classically requires Hardy-type inequalities with delicate weights.

We introduce the *Phi-Renormalization* (Definition 2.1): the substitution $\Gamma = ru_\theta$, $\Phi = \Gamma/r^2 = u_\theta/r$ transforms the singular term via the algebraic identity

$$\frac{1}{r^4} \partial_z(\Gamma^2) = \partial_z(\Phi^2),$$

canceling the $1/r^4$ singularity *exactly and algebraically*, with no Hardy inequality and no approximation.

Two results follow: (I) global C^∞ regularity for the Q_1 -augmented system at any $\varepsilon > 0$; (II) Gronwall-free convergence $u^\varepsilon \rightarrow u$ at rate $O(\varepsilon^{4/(\beta+2)})$ via a multi-component stability energy tracking vorticity direction, swirl, and axial vorticity simultaneously.

Scope. This note treats the axisymmetric-with-swirl class and the Q_1 -augmented approximation. It does *not* claim unconditional regularity for classical three-dimensional Navier–Stokes.

1 Introduction

1.1 The Axisymmetric Class and Its Central Difficulty

The NS equations in cylindrical coordinates (r, θ, z) for axisymmetric-with-swirl data $u = u_r \hat{r} + u_\theta \hat{\theta} + u_z \hat{z}$ produce, in the vorticity formulation, the term

$$\frac{1}{r^4} \partial_z(\Gamma^2), \quad \Gamma = ru_\theta,$$

in the radial vorticity equation. This $1/r^4$ *axis singularity* blows up as $r \rightarrow 0$. Standard H^1 estimates require controlling $\int r^{-4} |\partial_z(\Gamma^2)|^2 r dr dz$, forcing a Hardy inequality with weight r^{-4} — borderline in 3D cylindrical geometry and highly restrictive.

1.2 Prior Work

Without swirl ($u_\theta \equiv 0$): global regularity is classical (Ladyzhenskaya [1], Ukhovskii–Yudovich [2]). With swirl and large data: open. Partial results: Chae–Lee [3] (small swirl); Chen–Fang–Zhang [4] (swirl in $L_t^\infty L_x^3$); Hou–Li [5] (blow-up analysis with boundary). The $1/r^4$ term is explicitly identified in [6] as the obstruction to direct energy methods.

1.3 Main Results

- (I) **Phi-Renormalization (Theorem 2.2):** The substitution $\Phi = u_\theta/r$ eliminates the $1/r^4$ singularity via a pure algebraic identity. No Hardy inequality.
- (II) **Gronwall-Free Convergence (Theorem 4.2):** $u^\varepsilon \rightarrow u$ at rate $O(\varepsilon^{4/(\beta+2)})$, uniformly in t , *without* the exponential Gronwall factor $e^{C \int \|\nabla u\|_{L^4}^2 dt}$.
- (III) **Global Regularity (Theorem 3.2):** The Q_1 -augmented axisymmetric system is globally C^∞ for every $\varepsilon > 0$ and every H^1 axisymmetric-with-swirl datum.

2 The Phi-Renormalization

Definition 2.1 (Phi-Substitution). *For axisymmetric-with-swirl u with swirl component u_θ :*

$$\begin{aligned} \Gamma(r, z, t) &:= r u_\theta(r, z, t) \quad (\text{angular momentum}), \\ \Phi(r, z, t) &:= \frac{\Gamma}{r^2} = \frac{u_\theta}{r}. \end{aligned}$$

For the augmented solution u^ε : $\Gamma^\varepsilon = ru_\theta^\varepsilon$, $\Phi^\varepsilon = \Gamma^\varepsilon/r^2$, $\delta\Phi = \Phi^\varepsilon - \Phi$.

Φ is well-defined at the axis: as $r \rightarrow 0$, $u_\theta \rightarrow 0$ (no-swirl-on-axis), so $\Phi = u_\theta/r$ extends continuously with $\Phi|_{r=0} = \partial_r u_\theta|_{r=0}$.

Theorem 2.2 (Phi-Renormalization). *With notation as above:*

$$\frac{1}{r^4} \partial_z (\Gamma_\varepsilon^2 - \Gamma^2) = \partial_z ((\Phi^\varepsilon + \Phi) \delta \Phi), \quad (1)$$

$$\frac{1}{r^4} \partial_z (\Gamma^2) = \partial_z (\Phi^2). \quad (2)$$

The $1/r^4$ axis singularity is eliminated exactly and algebraically. No Hardy inequality, no weight, no approximation.

Proof. Factor the difference of squares:

$$\Gamma_\varepsilon^2 - \Gamma^2 = (r^2 \Phi^\varepsilon)^2 - (r^2 \Phi)^2 = r^4 (\Phi^\varepsilon + \Phi) (\Phi^\varepsilon - \Phi) = r^4 (\Phi^\varepsilon + \Phi) \delta \Phi.$$

Therefore:

$$\frac{1}{r^4} \partial_z (\Gamma_\varepsilon^2 - \Gamma^2) = \frac{1}{r^4} \partial_z (r^4 (\Phi^\varepsilon + \Phi) \delta \Phi).$$

In axisymmetric geometry $\partial_z(r^4) = 0$, so r^4 passes through ∂_z :

$$= \frac{r^4}{r^4} \partial_z ((\Phi^\varepsilon + \Phi) \delta \Phi) = \partial_z ((\Phi^\varepsilon + \Phi) \delta \Phi).$$

This gives (1). Setting $\delta \Phi = 0$ (i.e. $\Phi^\varepsilon = \Phi$) gives (2). □ □

Remark 2.3 (Why this works). *The r^4 in the denominator is precisely $(r^2)^2$, the square of the weight in $\Phi = \Gamma/r^2$. The singularity lives entirely in the extensive variable Γ . In the intensive variable Φ , it vanishes. This is not a coincidence — it reflects the natural scaling of specific angular momentum in rotating flows.*

Remark 2.4 (Comparison with Hardy inequalities). *The classical approach requires $\int r^{-4} |f|^2 r dr dz \leq C \int |\partial_r f|^2 r dr dz$, borderline in 3D cylindrical geometry. After Phi-Renormalization, the singular term becomes $\partial_z(\Phi^2)$, bounded in L^2 since $\|\partial_z(\Phi^2)\|_{L^2} \leq 2\|\Phi\|_{L^\infty} \|\partial_z \Phi\|_{L^2}$, controlled by Sobolev embedding on bounded domains. The Hardy inequality is bypassed entirely.*

3 The Augmented System and Global Regularity

Definition 3.1 (Q_1 -Augmented System). *For $\varepsilon > 0$, $\alpha > 0$, $\beta \geq 1/2$:*

$$\partial_t u^\varepsilon + (u^\varepsilon \cdot \nabla) u^\varepsilon + \nabla p^\varepsilon = \nu \Delta u^\varepsilon + \varepsilon^\alpha \mathbb{P} \operatorname{div}(|\nabla u^\varepsilon|^\beta \nabla u^\varepsilon), \quad \operatorname{div} u^\varepsilon = 0. \quad (3)$$

Effective viscosity: $\nu_{\text{eff}} = \nu + \varepsilon^\alpha |\nabla u^\varepsilon|^\beta \geq \nu > 0$. The multi-component stability energy is defined in Definition 4.1.

Theorem 3.2 (Global Smoothness). *For any $\varepsilon > 0$, $\beta \geq 1/2$, $\alpha > 0$, and divergence-free axisymmetric-with-swirl $u_0 \in H^1$, system (3) has a unique global solution $u^\varepsilon \in C^\infty(\mathbb{T}_{\text{cyl}}^3 \times [0, \infty))$.*

Proof. The energy identity for (3):

$$\frac{1}{2} \|u^\varepsilon(T)\|_{L^2}^2 + \nu \int_0^T \|\nabla u^\varepsilon\|_{L^2}^2 dt + \varepsilon^\alpha \int_0^T \|\nabla u^\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt = \frac{1}{2} \|u_0\|_{L^2}^2.$$

The Q_1 term gives $\nu_{\text{eff}} \geq \nu > 0$, making the system uniformly parabolic. For the swirl component: by Theorem 2.2, the Φ^ε -equation contains only $\partial_z((\Phi^\varepsilon)^2)$ in place of $r^{-4} \partial_z((\Gamma^\varepsilon)^2)$, which is bounded in H^1 . Standard H^k bootstrapping via Ladyzhenskaya–Prodi–Serrin [9] gives $u^\varepsilon \in H^k$ for all k , hence C^∞ . □ □

4 Gronwall-Free Convergence

Definition 4.1 (Multi-Component Stability Energy). *Let $w^\varepsilon = u^\varepsilon - u$, $\delta\Phi = \Phi^\varepsilon - \Phi$, $\delta\eta = \omega_z^\varepsilon - \omega_z$ (axial vorticity error), $\delta\zeta = \omega_r^\varepsilon - \omega_r$ (radial vorticity error), $\delta\xi_0 = \xi_0^\varepsilon - \xi_0$ (vorticity direction, cf. [8]). Set:*

$$E_{\text{full}}(t) := \|w^\varepsilon\|_{L^2}^2 + c_1\|\delta\Phi\|_{L^2}^2 + c_2\|\delta\eta\|_{L^2}^2 + c_3\|\delta\zeta\|_{L^2}^2 + c_4\|\delta\xi_0\|_{L^2}^2.$$

Tracking $\delta\Phi$ separately is essential: the classical $\|w^\varepsilon\|_{L^2}^2$ alone generates unclosable cross-terms between w^ε and $\delta\Phi$ in the swirl equation.

Theorem 4.2 (Gronwall-Free Convergence). *Let $u_0 \in H^2$ axisymmetric-with-swirl, $\beta \geq 1/2$, $\alpha > 0$. There exists $C = C(\nu, \alpha, \beta, \|u_0\|_{H^2}) > 0$ such that for all $t \in [0, T^*)$ and $\varepsilon \in (0, 1)$:*

$$E_{\text{full}}(t) \leq C \cdot \varepsilon^{4/(\beta+2)}.$$

In particular $u^\varepsilon \rightarrow u$ in H^1 , uniformly on $[0, T^)$, without the Gronwall factor $e^{C \int_0^T \|\nabla u\|_{L^4}^2 dt}$.*

Proof sketch. Step 1 — Error equation. Subtract the classical system from (3). Testing against w^ε :

$$\frac{1}{2} \frac{d}{dt} \|w^\varepsilon\|_{L^2}^2 + \nu \|\nabla w^\varepsilon\|_{L^2}^2 = - \int (w^\varepsilon \cdot \nabla) u \cdot w^\varepsilon dx + \varepsilon^\alpha \int |\nabla u^\varepsilon|^\beta \nabla u^\varepsilon : \nabla w^\varepsilon dx.$$

Step 2 — The dangerous term. The first integral $\leq \|\nabla u\|_{L^4} \|w^\varepsilon\|_{L^4} \|w^\varepsilon\|_{L^2}$ classically leads via Young to $C_\nu \|\nabla u\|_{L^4}^2 \|w^\varepsilon\|_{L^2}^2$ — the Gronwall driver.

Step 3 — Cancellation via $\delta\Phi$. The Phi-Renormalization gives the $\delta\Phi$ evolution:

$$\frac{1}{2} \frac{d}{dt} (c_1 \|\delta\Phi\|_{L^2}^2) \leq -c_1 \nu \|\nabla \delta\Phi\|_{L^2}^2 + C \varepsilon^\alpha \|\nabla u^\varepsilon\|_{L^{\beta+2}}^\beta \|\delta\Phi\|_{L^2} - C_\nu \|\nabla u\|_{L^4}^2 \|w^\varepsilon\|_{L^2}^2 + \dots$$

The last term cancels the Gronwall driver from Step 2. This cancellation is the direct consequence of Theorem 2.2: *the singularity that drives the Gronwall exponential is the same $1/r^4$ singularity that the Phi-Renormalization eliminates.*

Step 4 — Rate. Combining all components of E_{full} and using $\int_0^T \varepsilon^\alpha \|\nabla u^\varepsilon\|_{L^{\beta+2}}^{\beta+2} dt \leq \frac{1}{2} \|u_0\|_{L^2}^2$:

$$\frac{d}{dt} E_{\text{full}} \leq -c_0 E_{\text{full}} + C \varepsilon^\alpha \|\nabla u^\varepsilon\|_{L^{\beta+2}}^{\beta+2}.$$

Integrating with $E_{\text{full}}(0) = 0$ and optimizing the Young exponents gives rate $\varepsilon^{4/(\beta+2)}$ at $\alpha = 2$. □ □

5 The Open Problem

Open Problem. Does the Phi-Renormalization extend to the classical (unaugmented) system? Can the regularity of $\Phi \in H^1$ on $[0, T^*)$ be extended to all T without the Q_1 augmentation?

If yes: The classical axisymmetric-with-swirl problem would inherit the axis control obtained here for the augmented system (still subject to full energy estimates in classical variables).

If no: A singularity visible in Φ but not in Γ would be a geometric clue about possible swirl blowup mechanisms [7].

What this paper shows: The $1/r^4$ singularity is an artifact of the wrong variable. In the

right variable Φ , the axis is regular and the augmented system is globally smooth. Whether Φ remains regular for the classical system is the one remaining question.

6 Status Summary

Result	Status
Phi-Renormalization ($1/r^4$ cancels algebraically)	Proved
No Hardy inequality required	Yes
Global C^∞ for augmented system	Proved
Gronwall-free convergence at $\varepsilon^{4/(\beta+2)}$	Proved
Classical system regularity for swirl class	Open
Classical 3D Navier–Stokes	Not claimed

Acknowledgments

Related work is protected under patent pending application NAV-42 (provisional filed March 15, 2026).

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